

MUHAMMAD BILAL

Data Analyst | Python | Power BI | Data Visualization

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PROFESSIONAL SUMMARY

BS Data Science student focused on data analysis, business intelligence, and practical data visualization. Experienced through hands-on projects in data cleaning, exploratory analysis, dashboard development, web-scraped datasets, and analytical reporting. Comfortable with Python, pandas, NumPy, Power BI, Matplotlib, Plotly, Excel, and Git/GitHub. Seeking a Data Analyst or Junior Data Analyst opportunity.

EDUCATION

Air University, Islamabad — BS Data Science
3rd Semester

Expected 2029

TECHNICAL SKILLS

Data Analysis: Python, pandas, NumPy, data cleaning, exploratory data analysis

Visualization & BI: Power BI, dashboard development, Matplotlib, Plotly

Business Analytics: Excel, KPI-oriented reporting, data visualization

Data Collection: Web scraping, BeautifulSoup, APIs

Tools: Git, GitHub, Jupyter Notebook, Streamlit

SELECTED PROJECTS

E-commerce Price Monitoring Scraper & Dashboard

Python, BeautifulSoup, Playwright, SQLAlchemy, SQLite/PostgreSQL, Streamlit, Plotly

- Built an automated pipeline to collect competitor product prices, availability, and historical price changes.
- Added data normalization, configurable CSS selectors, change detection, CSV reporting, and Slack/email alerts.
- Developed an interactive Streamlit dashboard for price comparison, trends, configuration, and alert history.

[View GitHub repository →](#)

CVD Health Prediction

Python, pandas, scikit-learn, Logistic Regression, EDA, data cleaning, Matplotlib

- Performed exploratory data analysis and preprocessing on a 308,854-row cardiovascular health survey dataset.
- Handled duplicates, categorical encoding, outliers, standardization, and class imbalance using class-weighted Logistic Regression.
- Evaluated the model using accuracy, precision, recall, F1, and confusion-matrix metrics.

[View GitHub repository →](#)

Text Classification Pipeline

Python, pandas, scikit-learn, TF-IDF, Logistic Regression, Naive Bayes, Matplotlib

- Built an end-to-end CSV → cleaned data → feature engineering → classification workflow for SMS spam detection.
- Compared Multinomial Naive Bayes and Logistic Regression using accuracy, precision, recall, F1, and a confusion matrix.
- Selected Logistic Regression, achieving 97.7% accuracy and a 0.908 F1-score on the spam class.

[View GitHub repository →](#)

CERTIFICATIONS

- Python Programming for All Levels — Pakistan Skill Development Initiative
- Data Analytics and Business Intelligence — DigiSkills.pk
- AI and Data Science for Beginners — Pakistan Skill Development Initiative
- AWS Foundations: Getting Started with the AWS Cloud Essentials — Amazon Web Services (AWS)